

Cold-Start Affordability for Pay-in-4 BNPL

Approve more thin-file customers while losing less to first-payment default

The problem

The thin-file pincer

- About half of Pay-in-4 applicants have little or no credit-bureau history
- Approve too loosely and you eat first-payment-default losses, plus FCA Consumer Duty affordability scrutiny
- Decline too hard and you lose good customers, GMV, and merchant relationships
- Static rules and stale bureau pulls do not reflect a thin-file customer's real ability to pay

Who feels it

Executive sponsor (VP Credit Risk / CFO)

Net credit loss

Approval rate and GMV

Customer LTV

Regulatory exposure

Domain owner (Head of Collections)

Analyst hours per cured account

Reason-code explainability

Cure success rate

Decision latency

The outcome

Proof: lower loss at the same approval rate

Policy	Approval rate	Approved FPD	Net loss (proxy)
Bureau-only baseline	85%	5.22%	48.7k GBP
FPD affordability model	85%	3.39%	32.3k GBP
Improvement	held constant	35% lower	35% lower

How it works: a closed loop

- Score first-payment-default risk at checkout from real-time alternative data
- Approve or decline against a per-cohort threshold, with deterministic reason codes
- When an account slips, GenAI drafts an affordability narrative and cure note for human review
- Genie surfaces the worst merchant cohort; the analyst tightens that threshold
- The next decision reads the new threshold, so the system learns

One integrated journey on Databricks

Stage	What it does here
Lakeflow	Ingest checkout, repayment, open-banking, device events; bronze to gold
Unity Catalog	PII masks, region row filter, sensitivity tags, full lineage
Lakebase	Live early-cure case queue and cohort thresholds the app writes
ML + GenAI	FPD model with SHAP reason codes; GenAI affordability narrative
Genie	Ask which cohort drives FPD; the answer drives a threshold action
Databricks App	Approval and Early-Cure Console for the risk team

ROI sensitivity model (illustrative)

- You set the inputs: annual GMV, baseline FPD rate, matched approval rate, recovery rate
- Illustrative anchor at 4B GBP GMV and 5.5% baseline FPD: roughly 35 to 45M GBP net loss avoided per year
- Plus GMV upside from cutting thin-file over-declines 8 to 12%
- Every figure is illustrative on synthetic data; it demonstrates the mechanism, not your portfolio

Governance and trust

Built-in controls

PII column masks and region row filter
Sensitivity tags and end-to-end lineage
Region excluded from the model for fairness

Decisions you can defend

Deterministic adverse-action reason codes
GenAI drafts for human review, guardrailed
Every decision traceable for Consumer Duty

Demo versus production (the honest slide)

- Synthetic data proves the mechanism and the architecture, not a portfolio number
- Scores are materialised to Lakebase; the console reads a decision as a fast key lookup
- GenAI output is a draft for human review, never auto-sent to customers
- Proposed pilot: 10k customers, 12-week holdout, matched approval rate, regulatory sign-off



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